CSE400 - Fundamentals of Probability in Computing

Lecture 3: Introduction to Probability Theory

Dhaval Patel, PhD

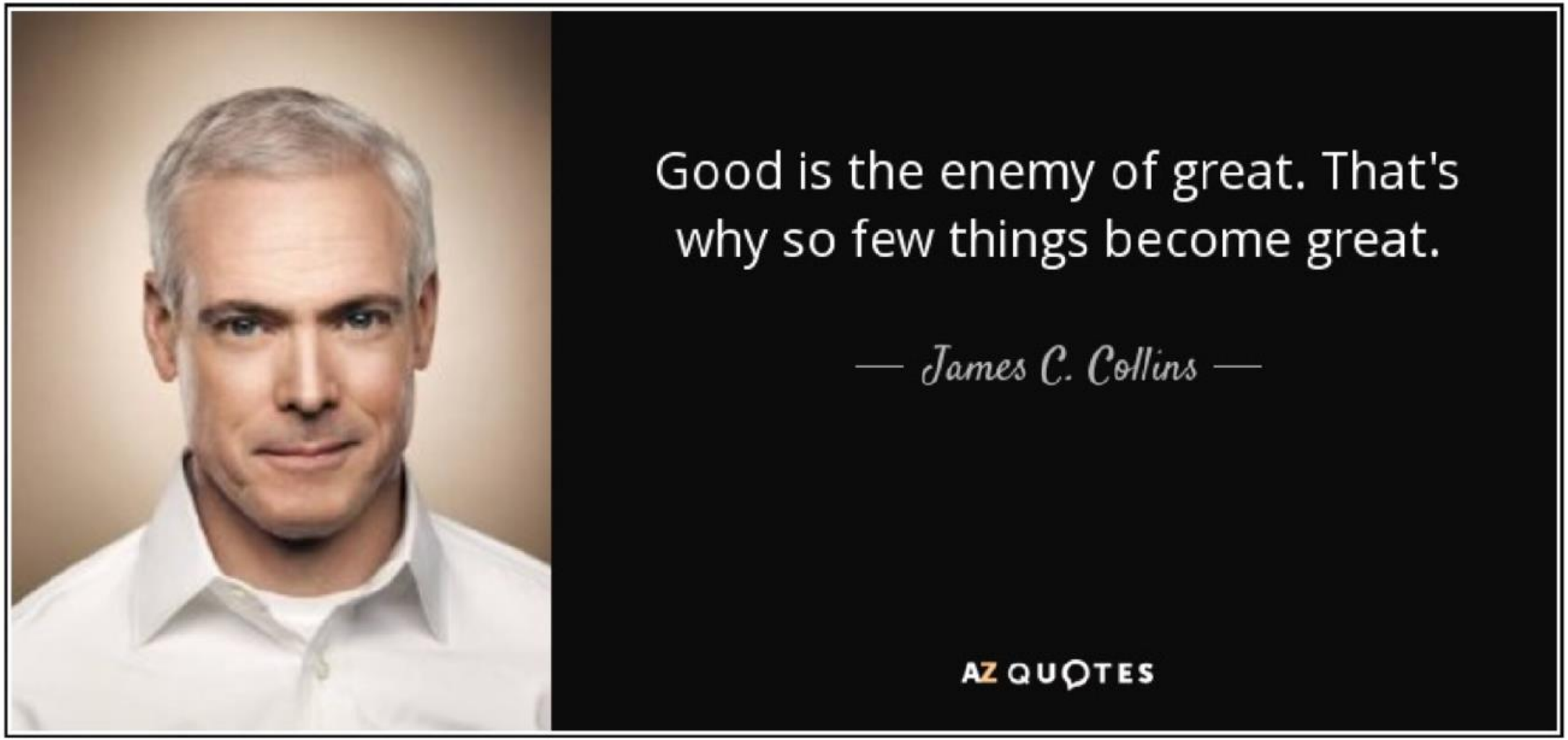
Associate Professor

Major Advisor - Computer Science and Engineering (CSE)

Machine Intelligence, Computing and xG Networks (MICxN) Research Lab

SEAS-Ahmedabad University, Ahmedabad, Gujarat- India

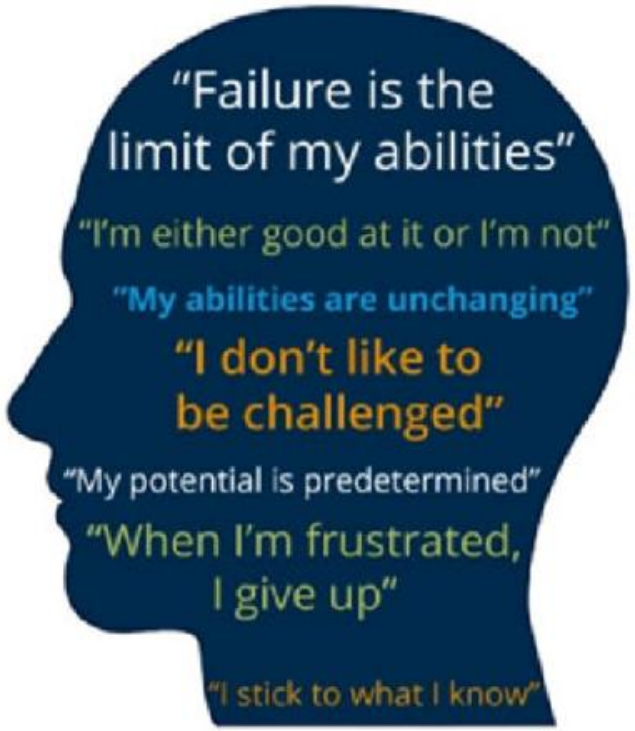
January 13, 2026


Good is the enemy of great. That's why so few things become great.
— James C. Collins —
AZ QUOTES


"Failure is an opportunity to grow"
"I like to try new things"
"I can learn to do anything I want"
"Challenges help me grow"
"My effort and attitude determine my abilities"
"Feedback is constructive"
"I am inspired by the success of others"
GROWTH
mindset



GROWTH mindset



FIXED mindset

Outline-1/2

- **CSE400: General Course Information**
  - Team
  - Active Learning Platform: Campuswire
  - Schedule
  - Grading

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  - Example: Daily life conversations

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- **Engineering Applications:**
  - Speech Recognition
  - System Radar System
  - Communication Network

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CSE400: General Course Information

- Team

Dr. Dhaval Patel

- Role: Instructor
- Place: Faculty Office (Room-210)
- Faculty Profile: https://ahduni.edu.in/seas/people/faculty/dhaval-patel
- Personal Webpage: http://profpatel.in/
- Area of Interest: xG Networks, Applied ML / DL / RL / AutoML, Intelligent Transportation Systems, Life Sciences, Behaviour Modelling using AI
- Email: dhaval.patel@ahduni.edu.in



CSE400: General Course Information

- Team



- **Deep Patel**
- BTech-CSE (3rd Year)
- Working in RL and Pinching Antenna Systems
- Email id: deep.p4@ahduni.edu.in



- **Prapti Patel**
- BTech-CSE (4th Year)
- Smart Sensing Framework Using Kolmogorov-Arnold Networks for Enhanced Detection in 5G Environments (COMSNETS'25)
- FAN-F: A Fourier Analysis Network-based framework for efficient and reliable Spectrum Sensing for 5G and Beyond (CCNC'26)
- Email id: prapti.p@ahduni.edu.in

CSE400: General Course Information

- Team



- **Raj Koticha**
- BTech-CSE (4th Year)
- Proactive Self-Aware Multi-Agent Reinforcement Learning for Resource Management in NR-V2X Platooning (COMSNETS'26)
- Email id: raj.k1@ahduni.edu.in



- **Ritu Patel**
- BTech-CSE (4th Year)
- Working in Intelligent Transportation System
- Email id: rituben.p@ahduni.edu.in

CSE400: General Course Information

- Team



- **Rushi Moliya**
- BTech-CSE (4th Year)
- Optimizing UAV Deployment for Enhanced Detection Performance in Multi-UAV Cooperative Sensing (NCC 2025).
- Optimizing Multi-UAV 3D Deployment for Energy-Efficient Sensing over Uneven Terrains (arXiv 2025).
- Email id: rushi.m@ahduni.edu.in



- **Ura Modi**
- BTech-CSE (3rd Year)
- Working in Pinching Antenna Systems
- Email id: ura.m@ahduni.edu.in

CSE400: General Course Information

- Active Learning and Class Discussion

CSE400: General Course Information

- Active Learning and Class Discussion

Course Website

- Section-1: https://campuswire.com/c/G20541104/
- Section-2: https://campuswire.com/c/G766840CE/

CSE400: General Course Information

- Active Learning and Class Discussion

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Campuswire!

Why use Campuswire?

1. To build your Confidence level: Participate anonymously, Post/Back-channel communication during lecture
2. Collaborative and Active Learning: Solve difficulties with online participation
3. Feedback on lecture session: Real-time feedback: Polling
4. Personal Communication with Instructor/TAs: DM - Direct Message communicator

CSE400: General Course Information

- Schedule

CSE400: General Course Information

- Schedule

Lecture Sessions

1. **Section-1**: 9:30 AM to 11:00 AM (Tuesday, Thursday) - GICT Room -136
2. **Section-2**: 1:00 PM to 2:30 PM (Tuesday, Thursday) - GICT Room -137

CSE400: General Course Information

- Schedule

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TA Hours: In person/Online: TAs

1. **Timings**: Yet to be finalized: A post will be made soon.

CSE400: General Course Information

- Connect with the Instructor

Timing for the Discussion and Difficulty Sessions: Instructor

1. Contact Hours: 24×7 through Campuswire
2. Best Practice: Put your queries on Campuswire!
3. Need more time: Feel free to DM - It's between student and instructor only - Private Conversation
4. External Engagement: UGRP-8: 2026 / Join with Offline Projects / Need Counseling / Cup of TEA for 10 Min: Write on office email ID.

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Important Note:

1. LaTeX Tutorial and Assignment Submission Guideline

CSE400: General Course Information

- Grading

- **Grading Policy:**
  - Class Participation: 5%
  - Homework Assignments: 20%
  - Project: 30%
  - Mid Semester Exam: 20%
  - End Semester Exam: 25%

CSE400: General Course Information

- Library Resources

Text Books

- Sheldon Ross, *First Course in Probability A*. Pearson Higher Ed, 2019.
- Mitzenmacher, Michael, and Eli Upfal, *Probability and computing: Randomization and probabilistic techniques in algorithms and data analysis*.
- Sheldon Ross, *Probability models for computer science*. San Diego: Harcourt Academic Press, 2002.

CSE400: General Course Information

- Library Resources(Continue)

Video Lectures

- https://nptel.ac.in/courses/117105085/
- http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-041-probabilistic-systems-analysis-and-applied-probability-fall-2010/video-lectures/

CSE400 – Project Kickoff (30%)

Project Team Formation



CSE400 Project Team Formation Activity (Winter 2026) - Team Member Details | Section 1

Dear students,

As discussed during the class, we are going to have a PROJECT component with a 25% assessment weight-age. The primary goal of the project is to do effective teamwork and achieve the defined key objectives for the selected project domain.

Deadline : 17th January 2026, Saturday – **EOD**

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Project Execution Guidelines

- **Weightage:** 30 %

- **Major Milestones to Achieve: M1-M6 (total 6 submissions throughout the semester) – 1 per Group**

  1. Concept Evolution Maps
  2. Scribe : Process & Decision-Making (Decision Logs (Why X over Y?) , Constraint, Alternatives considered, Final decision, Evidence used, Trade-off Matrices Cost vs. performance vs. risk tables)
  3. Multimodal Artifacts (Video / Audio / Visual): Think-Aloud Videos / One-Minute Insight Videos / Project Demo
  4. Question-Driven Artifacts: Probing depth of understanding
  5. Collaboration & Team Dynamics Artifacts

- **Deliverables:** Codes, Reports, Videos, etc. as and when specified
- **Team Assessment:** TEAMMATES - Before and After Midsemester
- **Project Viva and Submission:** Towards the end of the course

How to kick start the project execution?

M1: Kickstart: Team Formation, Area Identification, Background, Motivation and Problem formulation

M2: Mathematical Modeling : Mathematical Modelling of the problem selected in ANY domain: (RV, PMF/PDF, CDF, Multi variate RVs, Joint PMF/PDF/CDF etc.)

M3: Coding: Simulation + Computation

M4: Inference: Choose, Understand, and code a randomized algorithm.

M5: Randomized Algorithms: Apply the randomized algorithm to the domain problem. Randomized Algorithms for the problem and Results (new inferences in comparison to deterministic algorithm)

M6: Derive Bounds +Analysis + Compile and submit your final deliverables





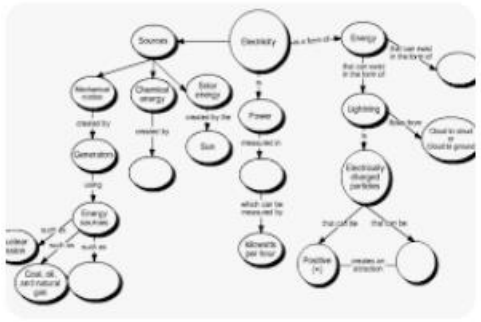




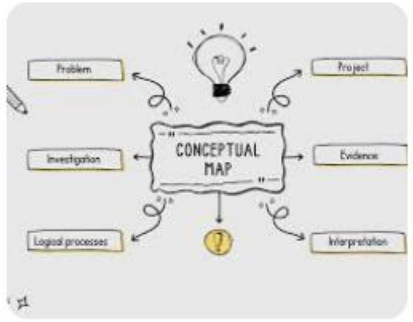


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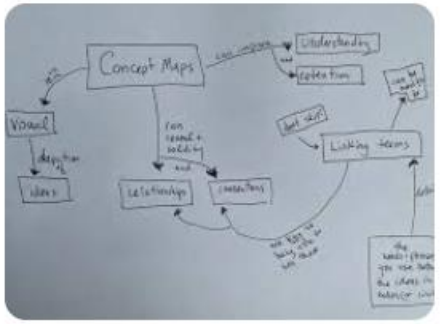
Submission #1 – Concept Evolution Maps



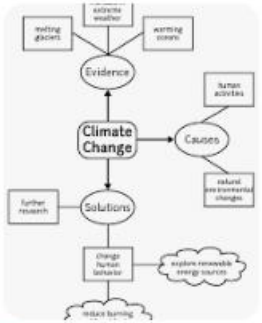
ResearchGate
Example concept map. | Download ...



Mindmap.Guide
Expert Concept Map Creation ...



Learning Strategies Center - Cornell ...
Concept Mapping – Learning Strat...

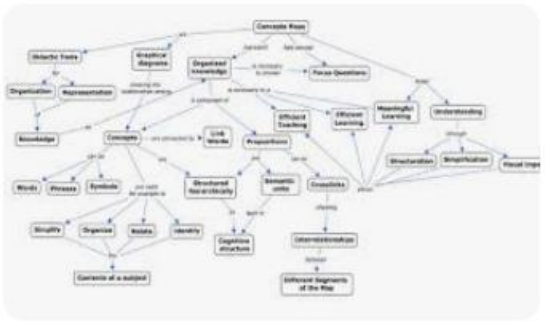


UNC Learning Cent...
Concept Maps – T...

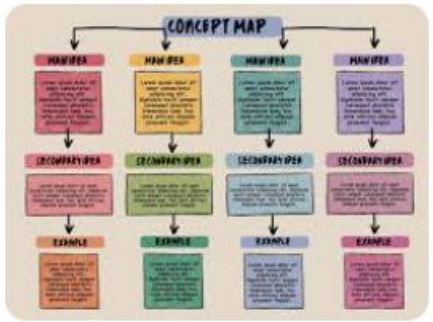


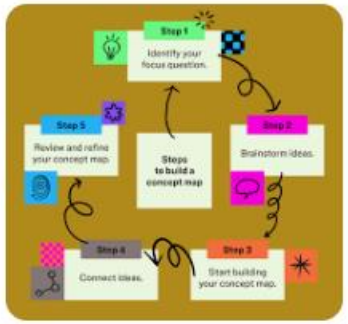
AdLit
Concept Maps | AdLit







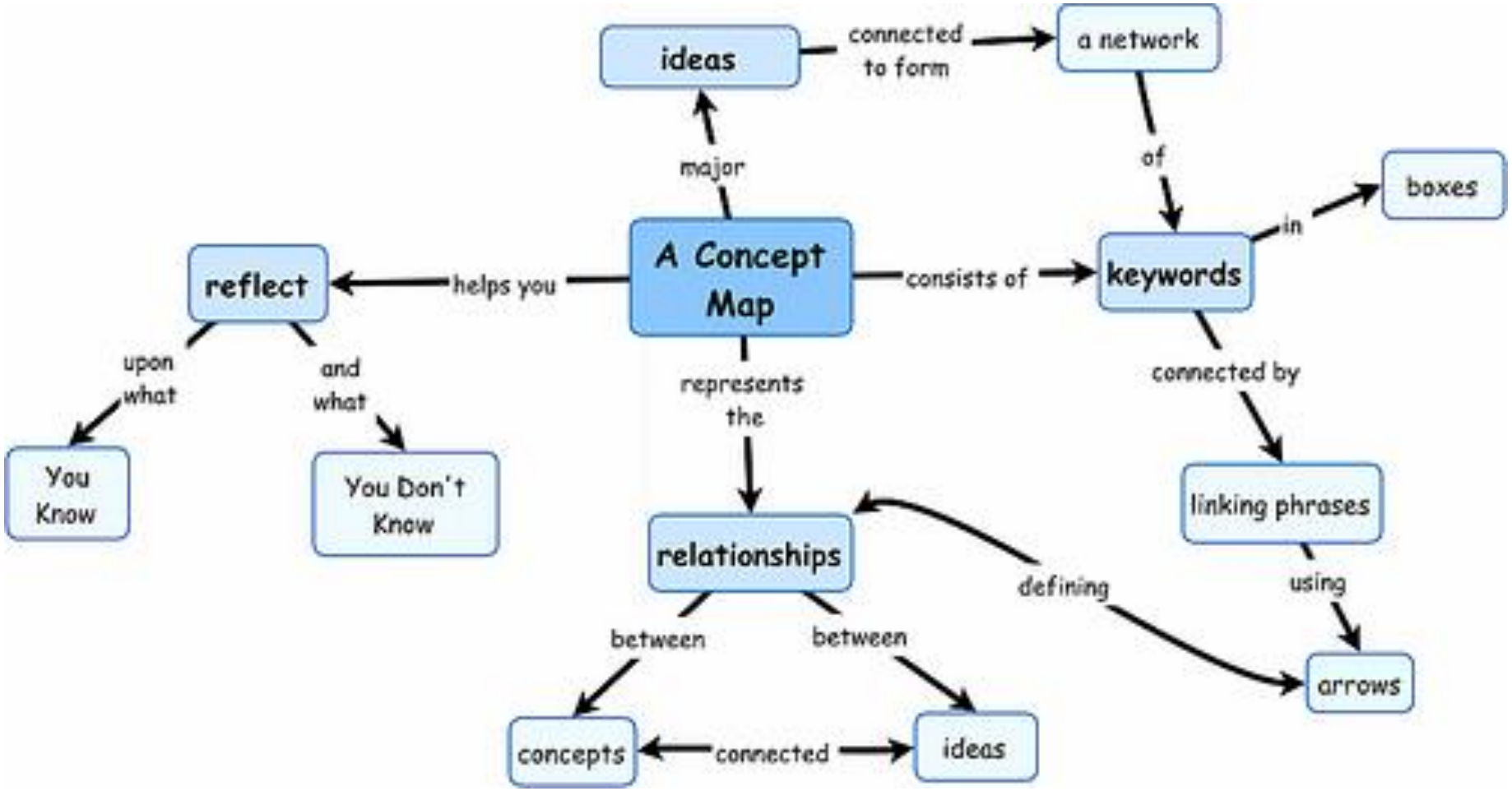




- **Tools:**
  - https://miro.com/concept-map/
  - https://app.diagrams.net/ (draw.io)

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Submission #1 – Concept Evolution Maps



Submission # 2 – Scribe : Learning Reflection Logs

- **Tools:**

  LaTeX **(Strictly)**

Types

(a) Lecture Scribes

- ✓ Every lecture – 2 groups will be assigned to generate a scribe which will reflect the lecture content with additional examples from the foriegn textbook(s) – **minimum 8-10 pages**

(b) Project Scribe

- ✓ Process & Decision-Making (Decision Logs (Why X over Y?) , Constraint, Alternatives considered, Final decision, Evidence used, Trade-off Matrices Cost vs. performance vs. risk tables)

- ✓ Total **6** submissions throughout the semester -**bi-weekly mode.** Each submission will focus on answering **specific questions** related to the project work.

Submission #3 – Multimodal Artifacts (Video / Audio / Visual):



Think-Aloud Videos / One-Minute Insight Videos / Project Demo

Prepare one video **per milestone**
~ 10-15 minutes

Explaining done in the corresponding **Milestone + Coding Simulation** (if applicable)

You may use PPT/Google Slides to narrate your work. Use any tool to record.

We are not looking for your editing skills, but the content you are delivering.

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Why should we learn CSE400?

- Conversations between Patient's Relative and Nurse

A patient is admitted to the hospital and a potentially life-saving drug is administered. The following dialog takes place between the nurse and a concerned relative.

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NURSE: I hope it works, we'll know tomorrow.

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(1) RELATIVE: Nurse, what is the probability that the drug will work?
NURSE: I hope it works, we'll know tomorrow.

(2) RELATIVE: Yes, but what is the probability that it will?
NURSE: Each case is different, we have to wait.

Why should we learn CSE400?

- Conversations between Patient's Relative and Nurse

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(2) RELATIVE: Yes, but what is the probability that it will?
NURSE: Each case is different, we have to wait.

(3) RELATIVE: But let's see, out of a hundred patients that are treated under similar conditions, how many times would you expect it to work?
NURSE (somewhat annoyed): I told you, every person is different, for some it works, for some it doesn't.

Why should we learn CSE400?

- Conversations between Patient's Relative and Nurse(Continue..)

(4) RELATIVE (insisting): Then tell me, if you had to bet whether it will work or not, which side of the bet would you take?
NURSE (cheering up for a moment): I'd bet it will work.

Why should we learn CSE400?

- Conversations between Patient's Relative and Nurse(Continue..)

(4) RELATIVE (insisting): Then tell me, if you had to bet whether it will work or not, which side of the bet would you take?
NURSE (cheering up for a moment): I'd bet it will work.

(5) RELATIVE (somewhat relieved): OK, now, would you be willing to lose two dollars if it doesn't work, and gain one dollar if it does?
NURSE (exasperated): What a sick thought! You are wasting my time!

Why should we learn CSE400?

- Conversational Analysis

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- Conversational Analysis

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NURSE: Each case is different, we have to wait.

In this conversation, the RELATIVE attempts to use the Concept of Probability to discuss an Uncertain Situation.

- The nurse's initial response indicates that the meaning of "probability" is not understood
- The RELATIVE tries to make it more clear

Why should we learn CSE400?

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- The approach is to define probability in terms of Frequency of Occurrence, as a % of successes in a moderately large number of similar situations.
- However, Nurse is not wrong:
  - What if this was an experimental drug that was administered for the very first time in this hospital or in the nurse's experience?

Why should we learn CSE400?

- Conversational Analysis

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- Conversational Analysis

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- Conversational Analysis

(4) RELATIVE (insisting): Then tell me, if you had to bet whether it will work or not, which side of the bet would you take?
NURSE (cheering up for a moment): I'd bet it will work.

- It is an attempt to infer the nurse's beliefs in an indirect manner.
- Since the nurse is willing to accept a one-for-one bet that the drug would work, we may infer that the Probability of Success is judged to be at least 50%.

Why should we learn CSE400?

Conversational analysis

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Conversational analysis

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Why should we learn CSE400?

Conversational analysis

(5) RELATIVE (somewhat relieved): OK, now, would you be willing to lose two dollars if it doesn't work, and gain one dollar if it does?

NURSE (exasperated): What a sick thought! You are wasting my time!

- If the nurse has accepted the last proposed bet (two-for-one), this would have a Probability of Success at least 2/3.

Why should we learn CSE400?

- A broad variety of contexts can be visualized
- Successful applications in science, Engineering, medicine etc
- Non-engineering applications: CASINO (Game of Chance), Insurance Industry, Meteorologist

Why should we learn CSE400?

Key Objectives and Outcomes:

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- To familiarize students with the basic axioms of probability and develop a strong foundation.

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- To introduce probabilistic bounds and their applications in algorithmic analysis, with a focus on Chernoff bounds and Hoeffding bounds.

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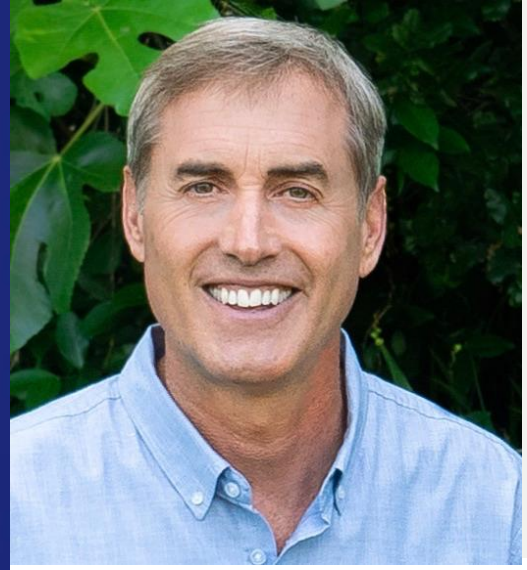
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- Important Engineering Applications:

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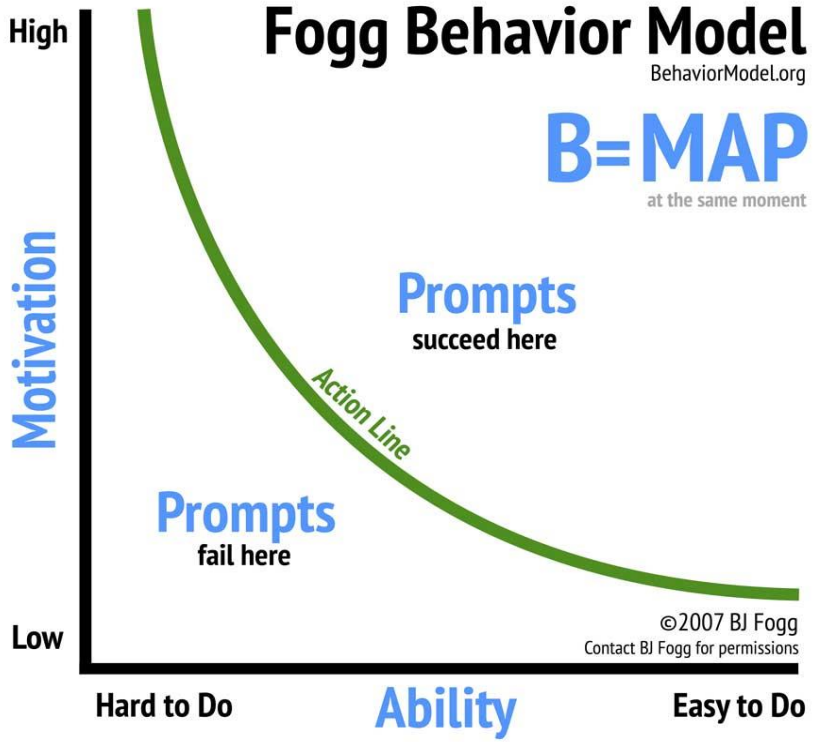
Introduction to UGRP-8 [2026 – 2027]

WHY IS RESEARCH IMPORTANT FOR UNDERGRADUATE STUDENTS?

- Strengthens critical thinking skills and analytical skills
- Increases Confidence
- Boosts Resume & Career Preparation
- Relationships with faculty mentors and peer researchers
- Being part of a on-campus scholarly network
- Opportunity to publish
- On-track to graduation
- Increases the probability of getting accepted in a graduate program

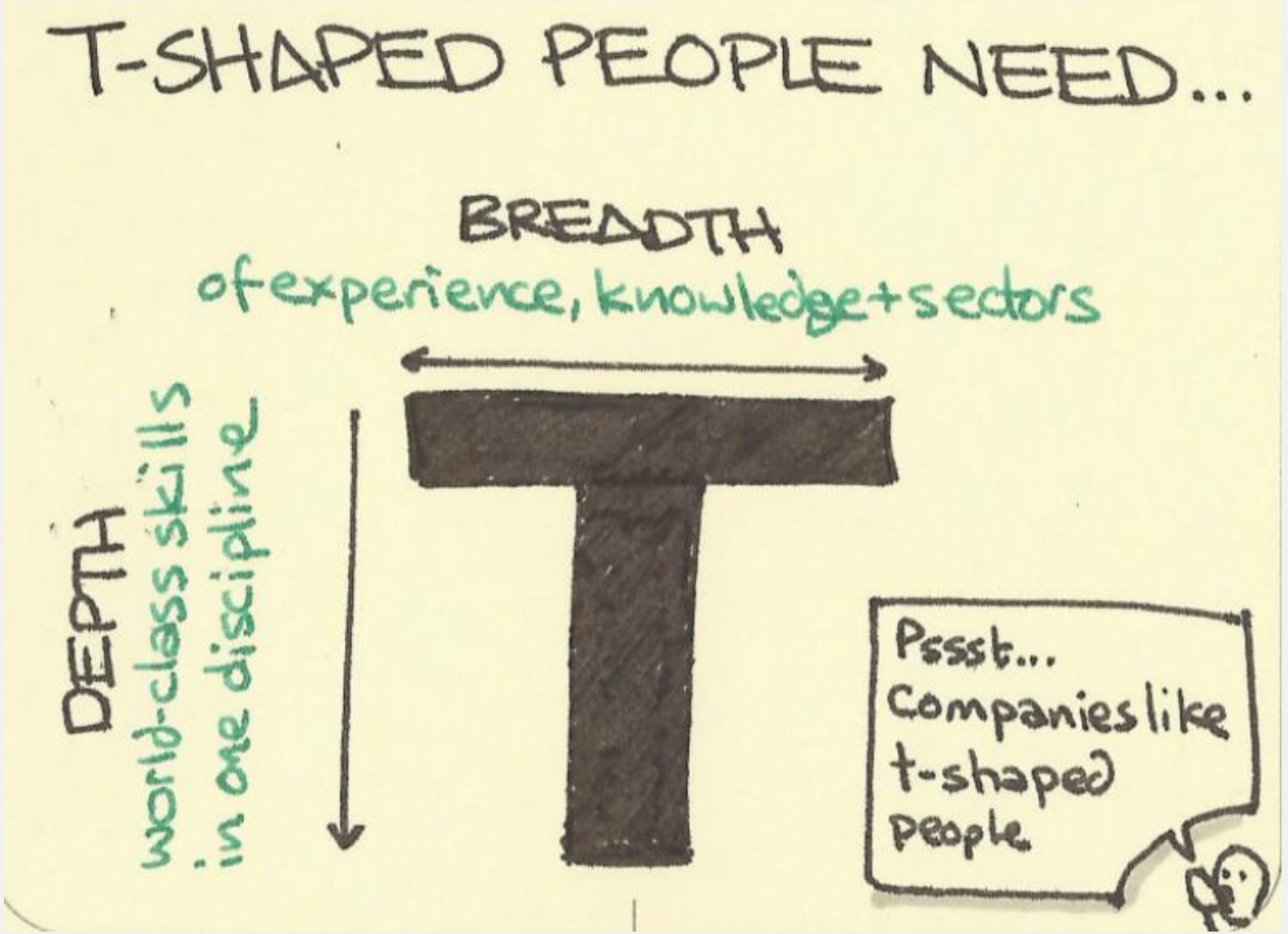


Dr. BJ Fogg founded the Behavior Design Lab at Stanford University, where he directs research and innovation.



Why UGRP?

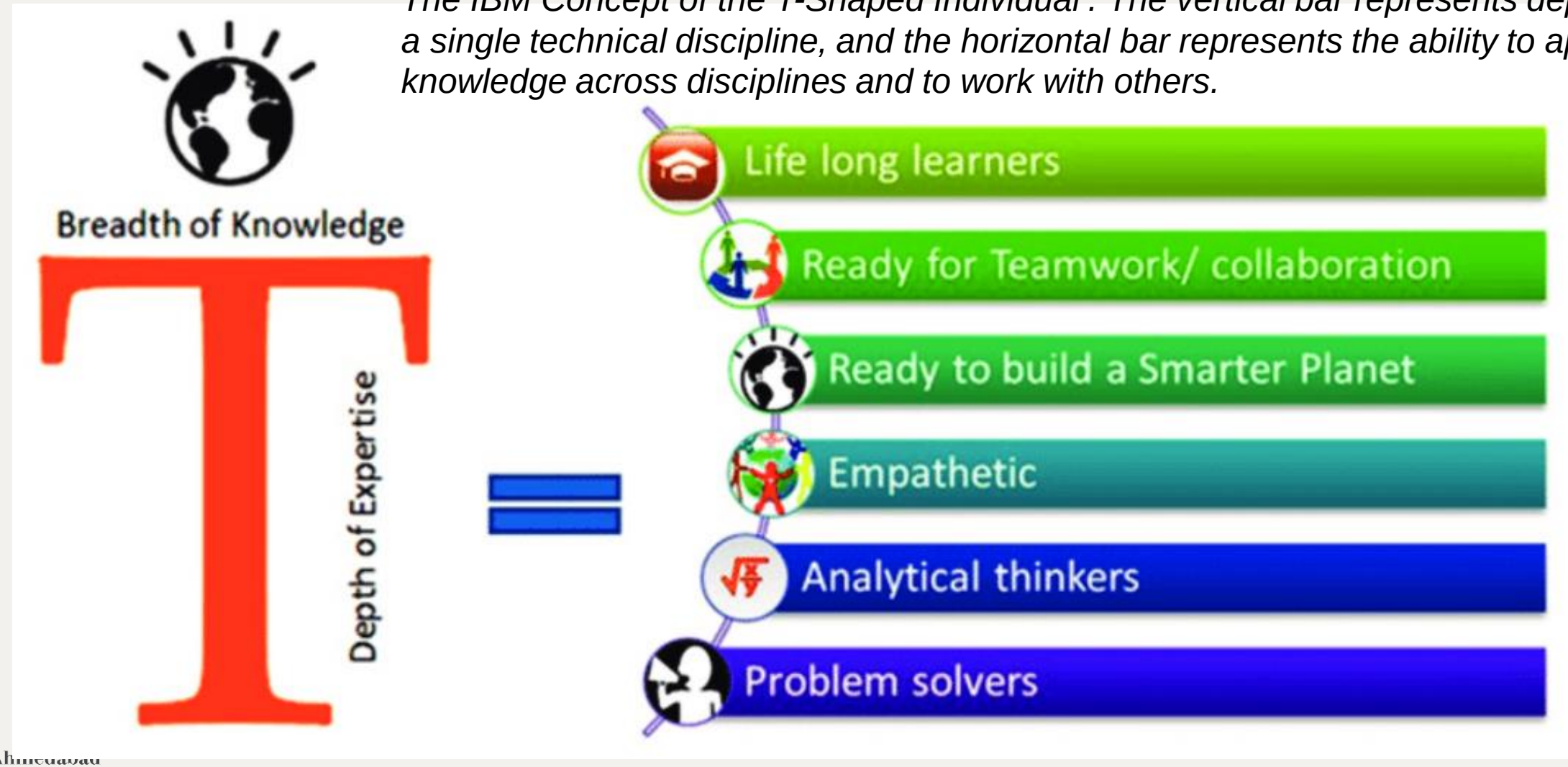
Tranquillo, Joe. "The t-shaped engineer." *Journal of Engineering Education Transformations* 30.4 (2017): 12-24.



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Why UGRP?

The IBM Concept of the T-Shaped Individual . The vertical bar represents depth in a single technical discipline, and the horizontal bar represents the ability to apply knowledge across disciplines and to work with others.



Philosophy for UGRP

Multidisciplinary:

Arts, Science and Management

Experiential Learning

Research Driven

4D Model:

Discover + Design + Develop + Deliver

Breadth:

CS and CSE

Data Science:

Applied Artificial Intelligence

Morden Computer System Design:
Hardware and Software

Networks

and

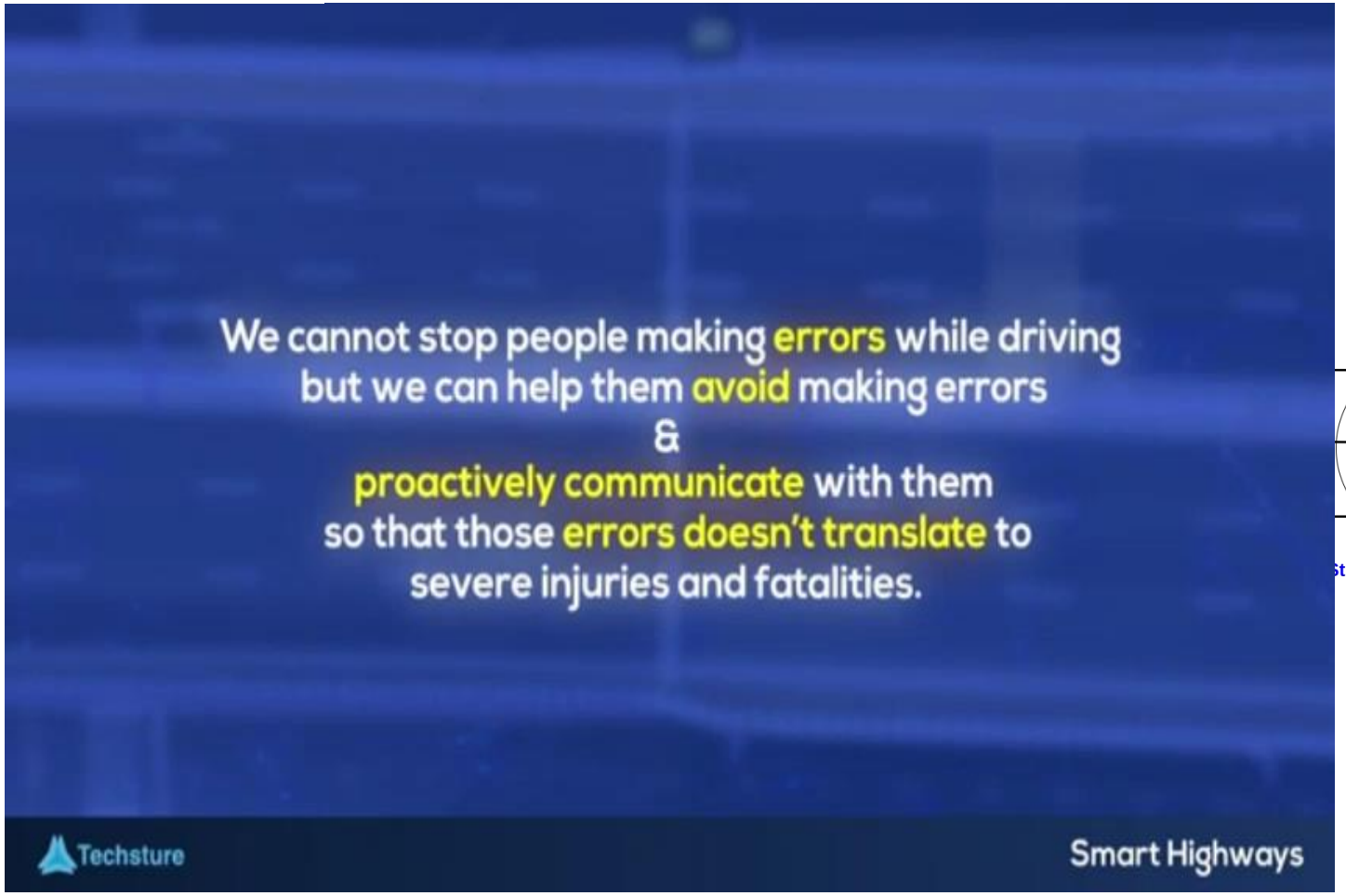
IoT/ IoBNT / IoV

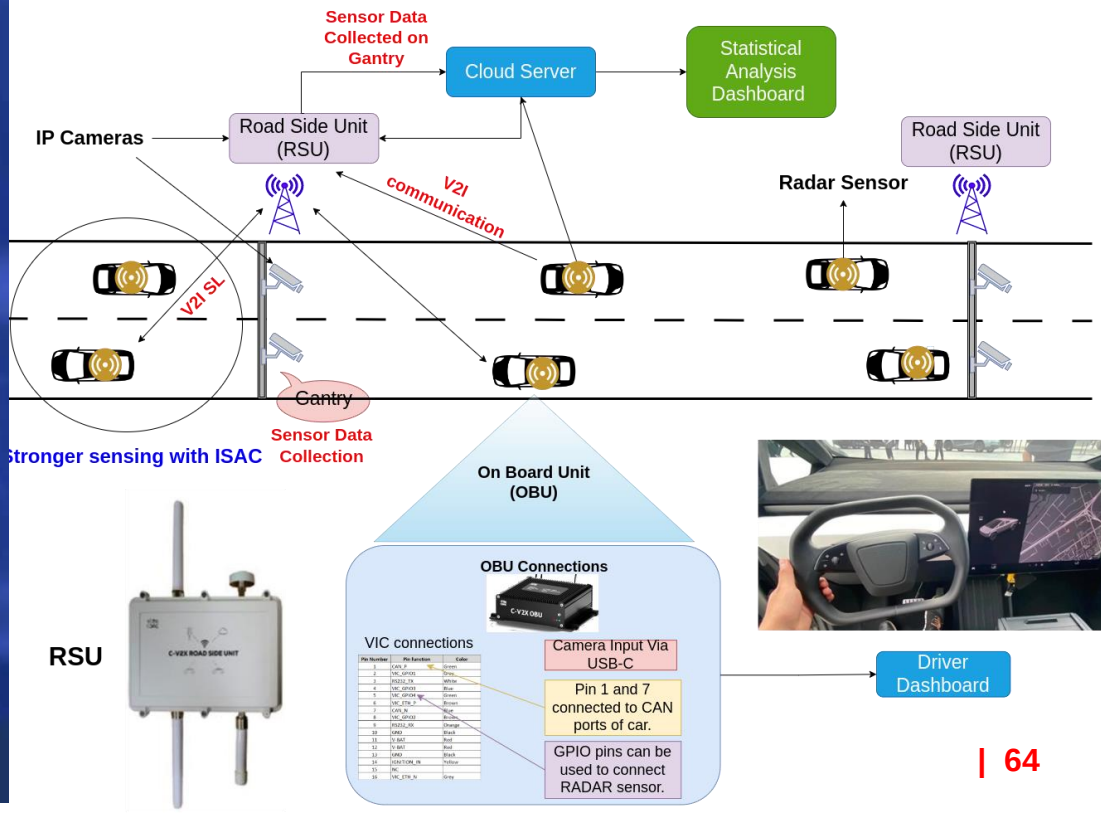
Current Activities – Industry

TechSture Technologies



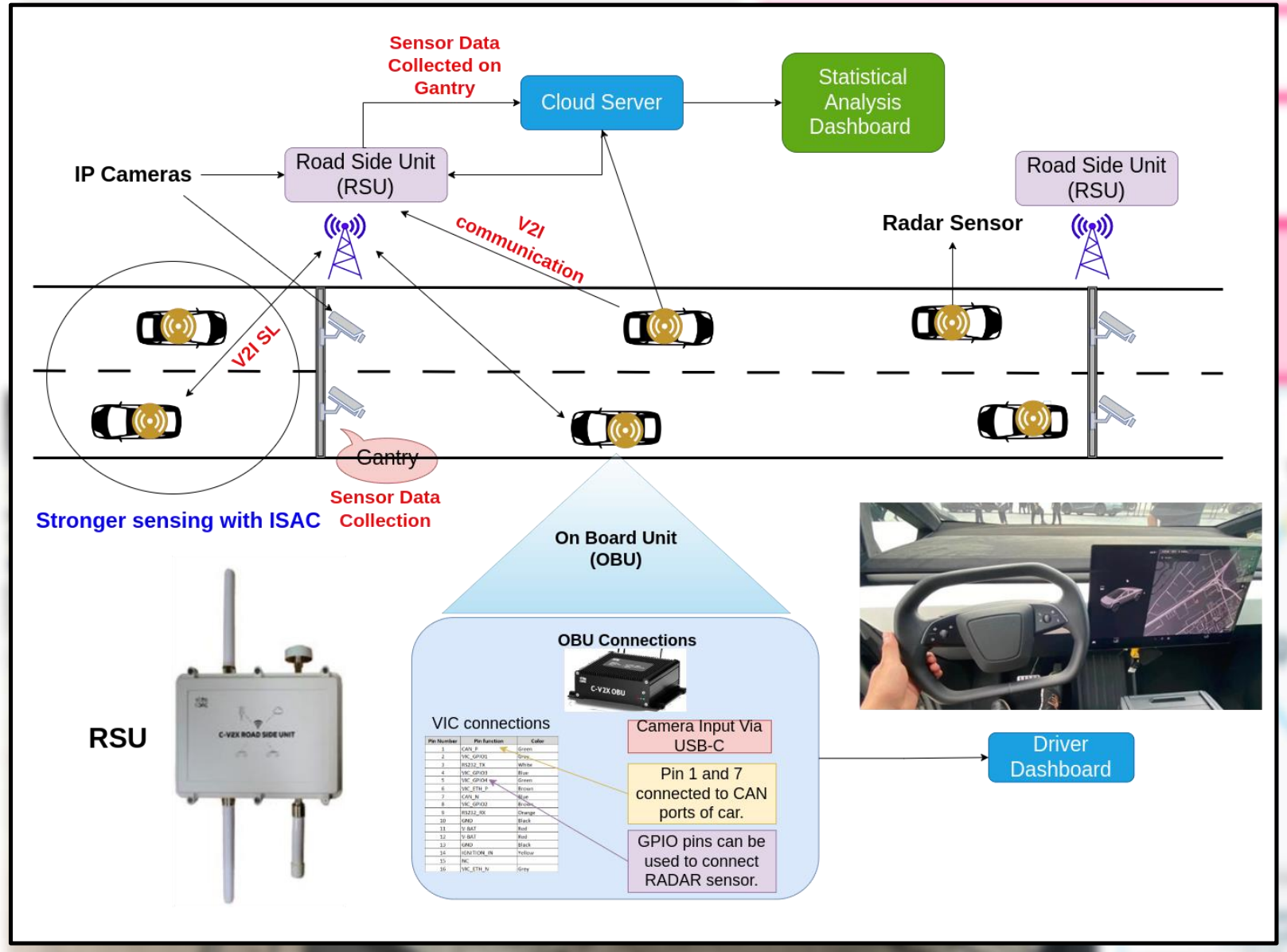






MICxN Lab – Research Project #2

Design and Development of 5G enabled Intelligent Transportation Systems Testbed in Gujarat



Submitted Proposal:

6G DEPARTMENT OF TELECOMMUNICATIONS
BHARAT 6G VISION

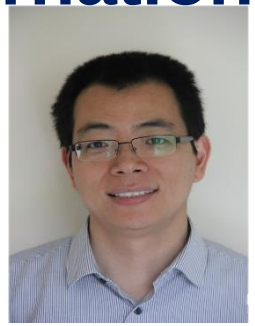
5G LAB

CFP: Accelerated Research on 6G Eco-System

"Deployment of 6G Accelerated Smart Transportation with C-V2X on Ahmedabad-Rajkot Highway (NH47) Corridor"

Industry-Academia Partnership

International Bilateral Collaborations



Zhiguo Ding
Professor in Communications, University of Manchester, UK



Miguel López-Benítez
Associate Professor, University of Liverpool, UK



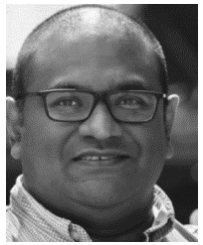
Guan Yong Liang
Professor of Communication Engineering, Nanyang Technological University, Singapore



Sumei Sun
Executive Director, Institute for Infocomm Research (A*STAR), Singapore



Bryan Ranger
Professor of Engineering, Boston College, USA



Siddhartan Govindasamy
Professor of Engineering, Boston College, USA



Janne J. Lehtomäki
Adjunct Professor, University of Oulu, Finland



Kenta Umebayashi
Professor, Tokyo University of Agriculture and Technology, Japan



Mazen Omar Hasna
Professor of Electrical Engineering, Qatar University



Joanne Mun-Yee Lim
Senior Lecturer, Monash University, Malaysia

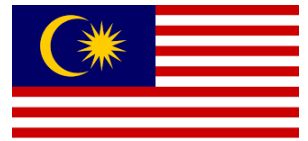


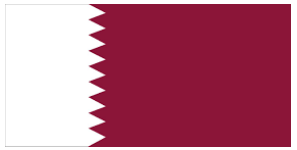
Ahmed Al-Tahmeesschi
Research Fellow, University of York, UK



Yoong Choon Chang
Associate Professor & Head - EEE Dept., Universiti Tunku Abdul Rahman (UTAR), Malaysia






MICxN Research Lab - Alumni



Yashraj Kakkad

Software Engineer,
Google



Nikhil Balwani

AI/ML Engineer,
ETS



Om Thakkar

AI Software Engineer,
Intel



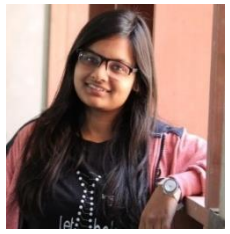
Maulin Vyas

ML Engineer,
Amazon



Bansi Gajera

Machine Learning Engineer,
GRUBBRR



Dhwani Mehta

Data Science Lead Assistant
Manager, EXL



Shivam Raval

NLP Data Scientist,
American Express



Deep Patel

Data Engineer,
Simplify Healthcare



Dhyey Thaker

Software Engineer, LexisNexis
Risk Solutions



Sanket Shah

Data Science Intern,
IQM Corporation



Shivam Thakker

Machine Learning Intern,
Survue



Hriday Nagrani

Research Aide,
Arizona State University

Google

ETS

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LexisNexis RISK SOLUTIONS

ASU Arizona State University

Simplify Healthcare

IQM

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MICxN Research Lab – Published Venues

IEEE Advancing Technology for Humanity
IEEE ComSoc IEEE Communications Society
IEEE Transactions on Communications
IEEE Transactions on Wireless Communications
IEEE Transactions on Vehicular Technologies
IEEE Transactions on NanoBioscience (TNB)
IEEE Transactions on Cognitive Communications and Networking

COMSNETS

NEC

IEEE INDICON

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Multidisciplinary | Rapid Review | Open Access Journal

IEEE pimrc

IEEE GLOBECOM®

CCNC

IEEE WCNC

Springer Nature Wireless Networks

CROWNCOM

IEEE TENSYMP

IET The Institution of Engineering and Technology
IET Communications Journal

IEEE COMMUNICATIONS LETTERS

IEEE VTS IEEE Vehicular Technology Society
Vehicular Technologies Conference

IEEE Signal Processing Society
International Conference on Signals and Systems (ICSigSys)

SPCOM International Conference on Signal Processing and Communications

CAMAN

22 nd International Conference on Machine Learning and Applications

Undergraduate Research Programme

Case study: Few UGRP Scholars at MICxN Research Lab

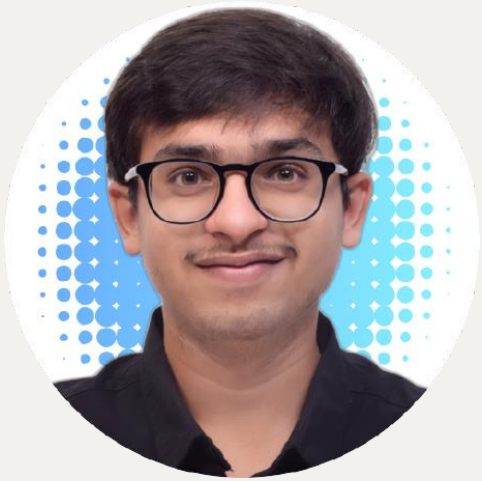


UGRP # 1 Maunil Vyas *(B.Tech ICT 2017 Passout)*
Published "ANN based hybrid spectrum sensing scheme for Cognitive" in IEEE PIMRC-2017, Montreal, Canada
Currently ML Engineer at Amazon!



UGRP # 2 : Nikhil Balwani
(B.Tech ICT 2019 Passout)
Published "LSTM based spectrum sensing scheme for Cognitive Radio" in PIMRC-2019, Istanbul, Turkey
Columbia Pass out, currently at Amazon

UGRP # 3:



Yashraj Kakkad
Coexistence of WiFi and LTE fo 5G
Acepted in
IEEE Transactions on Vehicular Technology
Software Engineer III at Google!

Jay Mohta

Ahmedabad University

Undergraduate Research Programme

Case study: Few UGRP Scholars

UGRP # 4 2021-2022 batch:



Abhishu Oza
Published his work in **IEEE Access**

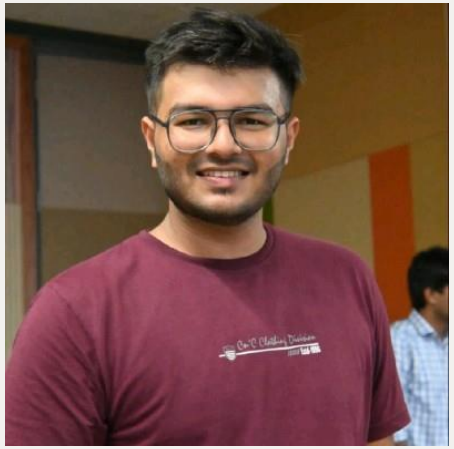


Omikumar Makadia
Published at **PIMRC**
currently **PhD at Rutgers**

UGRP # 5 2022-2023 batch:



Raj Dave
Currently at Northwestern University



Ayush Patel
Currently at Northeastern University

Undergraduate Research Programme

Venues: International Conferences



Undergraduate Research Programme

Venues: International Conferences



Undergraduate Research Programme

Venues: International Conferences



2 papers accepted in the main conference!

13 **Proactive Self-Aware Multi-Agent Reinforcement Learning for Resource Management in NR-V2X Platooning**

Raj Koticha (Ahmedabad University), Dhaval Patel (Ahmedabad University), Kirtan Kalaria (Ahmedabad University), Harsh Patel (Techsture Technologies India Pvt. Ltd.)

→ **UGRP-6**

24 **Curiosity-Driven Deep-RL for Adaptive Handover Optimization in O-RAN on the NH-47 Corridor**

Devashree Parikh (Ahmedabad University), Dhaval Patel (Ahmedabad University), Kashish Shah (Ahmedabad University), Harsh Patel (Techsture Technologies India Pvt. Ltd.)

→ **UGRP-7**

Best UG Presentation Award ←

Ahmedabad University

Undergraduate Research Programme

Venues: International Conferences



4 papers accepted in the main conference!

Best Paper Award ←

18 **Efficient Human Activity Recognition using Deep Unfolding in 802.11bf Wi-Fi Sensing**
Raj Dave (Ahmedabad University), Dhaval K Patel (Ahmedabad University)

Best UG Presentation Award ←

19 **Improving VR Performance in WiFi-6 using Deep Reinforcement Learning based Spatial Reuse**
Ayush Patel (Ahmedabad University), Dhaval K Patel (Ahmedabad University)

→ **UGRP-5**

19 **Optimized UAV Trajectory Design within Multi-Charging Station Scenario in the ISAC Framework**
Kunj Kanzariya (Ahmedabad University), Dhaval K Patel (Ahmedabad University)

20 **Smart Sensing Framework Using Kolmogorov-Arnold Networks for Enhanced Detection in 5G Environments**
Prapti Patel (Ahmedabad University), Dhaval Patel (Ahmedabad University)

→ **UGRP-6**

Ahmedabad University

Undergraduate Research Programme

Research outcomes from the MICxN Lab

COMSNETS'25

2025 17th International Conference on COMmunication Systems & NETworkS (COMSNETS)

Efficient Human Activity Recognition using Deep Unfolding in 802.11bf Wi-Fi Sensing

Raj N. Dave, *Student Member, IEEE* and Dhaval K. Patel, *Member, IEEE*
School of Engineering and Applied Science, Ahmedabad University
Ahmedabad, Gujarat, India - 380009
email:{raj.d@ahduni.edu.in, dhaval.patel@ahduni.edu.in}

979-8-3315-3119-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/COMSNETS63942.2025.10885555

***Abstract*—Human Activity Recognition (HAR) has emerged as a critical component for a diverse range of applications like smart homes, healthcare, security, etc. This paper proposes DUET-HAR a novel framework for HAR utilizing Deep Unfolding techniques to enhance Wi-Fi-based sensing capabilities. The framework addresses the limitations of current state-of-the-art learning-based implementations by integrating iterative optimization-inspired layers, enabling more efficient extraction of activity-related features from Channel State Information (CSI) signals. Our Deep Unfolding approach improves upon previous works by leveraging the interpretability and structured optimization of unfolding architectures, leading to enhanced accuracy and robustness. Our results demonstrate superior performance in terms of computational efficiency and accuracy when compared to state-of-the-art methods. We achieve a 50%, 90.6 % and 92.7 % reduction in FLOPs relative to LSTM, BLSTM and CSI-ResNeXt approaches, respectively. This improvement highlights the potential of Deep Unfolding to improve the efficiency of current HAR systems, making them more accurate and reliable.**

***Index Terms*—Human activity recognition, Wi-Fi sensing, Deep Unfolding, IEEE 802.11bf.**

prompted the exploration of more versatile and unobtrusive technologies for HAR, such as Wi-Fi-based HAR.

Wi-Fi-based HAR has numerous applications across industries, such as healthcare, security, and smart home automation. In healthcare, Wi-Fi-based HAR allows for continuous, non-intrusive monitoring of elderly or patients, enabling real-time detection of falls, inactivity, or distress. All of this is achieved without the need for wearable devices, thereby improving response times and safety [8] [9]. Similarly, HAR systems in smart homes can automatically adjust environmental controls like lighting, temperature, and appliances based on detected human activity, improving comfort and power efficiency [8]. Additionally, in security and surveillance, Wi-Fi-based HAR can detect unauthorized movements and suspicious behaviors, providing a cost-effective and easily deployable system for both residential and public settings [8].

Wi-Fi networks are already widely deployed in most residential, commercial, and public spaces, making them ubiqui-

COMSNETS'25

2025 17th International Conference on COMmunication Systems & NETworkS (COMSNETS)

Optimized UAV Trajectory Design within Multi-Charging Station Scenario in the ISAC Framework

Kunj Kanzariya, *Student Member, IEEE* and Dhaval K. Patel, *Member, IEEE*
School of Engineering and Applied Science, Ahmedabad University
Ahmedabad, Gujarat, India - 380009
email:{kunj.k1@ahduni.edu.in, dhaval.patel@ahduni.edu.in}

(COMSNETS) | 979-8-3315-3119-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/COMSNETS63942.2025.10885757

***Abstract*—This study explores integrated sensing and communication (ISAC) in the airborne domain, utilizing Unmanned Aerial Vehicles (UAVs) as dual-function communication base stations and radars. We propose a comprehensive methodology for jointly optimizing communication and sensing performances through optimized UAV trajectory design and energy-aware operations within scenarios containing multiple charging stations. Our approach used a multi-stage trajectory design with real-time battery management and Sequential Quadratic Programming based optimization. The average communication data rate and Cramer-Rao bound of mean-square error are considered for deriving communication and sensing performance of optimized trajectories. We formulate a normalized tradeoff objective as a weighted sum optimization problem. An iterative algorithm obtains locally optimal UAV trajectories. Numerical results demonstrate how UAV paths are determined by the balance between communication and sensing objectives and energy constraints. Comparisons with single-function UAV scenarios highlight improvements in overall mission efficiency, showcasing the benefits of the ISAC-based approach.**

***Index Terms*—Optimization, Sequential Quadratic Programming, UAV, Trajectory Design, Cramer-Rao bound, Constraints, Simulations**

for indoor localization, human movement classification and environment recognition [4], [5]. Traffic management, pedestrian safety and vehicle tracking are achieved by incorporating ISAC into intelligent transportation systems [6]. By unifying these functions, ISAC systems can deliver more efficient services. Static ISAC base stations suffer from the problem of the limited range, which affects their sensing and communication performances. UAV-based networks provide solutions to the drawbacks of static base stations. UAV's ability to provide flexible network structures and portable and on-demand connectivity, with a better range of operation, makes them particularly useful in emergency situations [7]–[9].

Moving UAVs offer better sensing and communication performances because of the capability of covering larger areas with Line-of-Sight dominant channels and getting rich, diverse positional information of sensory targets, which helps for better sensing [10]. While UAVs outperform stationary base stations, an unoptimized flight can lead to unfair resource allocation, inefficient battery usage and increased time consumption. To address these issues, researchers have mathe-

Undergraduate Research Programme

Research outcomes from the MICxN Lab

COMSNETS'25

2025 17th International Conference on COMmunication Systems & NETworkS (COMSNETS)

Smart Sensing Framework Using Kolmogorov-Arnold Networks for Enhanced Detection in 5G Environments

Prapti Patel, *Student Member, IEEE* and Dhaval K. Patel, *Member, IEEE*
School of Engineering and Applied Science, Ahmedabad University
Ahmedabad, Gujarat, India - 380009
email:{prapti.p@ahduni.edu.in, dhaval.patel@ahduni.edu.in}

works (COMSNETS) | 979-8-3315-3119-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/COMSNETS63942.2025.10885557

***Abstract*—Spectrum sensing is crucial for dynamic spectrum access in Cognitive Radio Networks (CRNs), particularly as we advance to 5G technology, which operates across higher frequency bands and supports ultra-reliable low latency communications. In this study, we propose a Kolmogorov-Arnold Network (KAN)-based spectrum sensing scheme specifically designed to address the challenges of low probability of detection in low Signal-to-Noise Ratio (SNR) environments commonly found in 5G networks. The proposed scheme improves detection performance by effectively handling the increased noise and interference associated with higher frequency bands and massive MIMO setups. Our experimental results demonstrate a significant improvement in the probability of detection at SNR = -10 dB. Compared to Long Short-Term Memory (LSTM) networks, and PU-DetNet, the KAN-based scheme shows an improvement in detection probability by 67.43%, and 47.29%, respectively. This enhancement is validated across diverse datasets, including empirical, 5G New Radio (NR), DeepSig, and radar datasets. The performance of the proposed KAN-based scheme is analyzed based on Accuracy, Precision, and F1-score which is crucial for optimizing spectrum utilization and ensuring reliable communications in the complex and high-demand 5G framework.**

***Index Terms*—Cognitive Radio, Advanced Spectrum Management, Kolmogorov-Arnold Networks (KAN), 5G Technology**

use the spectrum better while protecting co-existing users. Research shows that CR systems can be a promising solution to the problem of spectrum scarcity. Additionally, CR can help reduce interference in different areas, such as space, frequency, and time. This is especially critical since 5G seeks to maximize the reuse of spatial spectrum. However, this requires developing reliable spectrum sensing mechanisms that allow dynamic and non-disruptive access. However, in order to fully exploit the promise of CR integration into 5G, several issues need to be resolved. To overcome those challenges, efficient spectrum sensing schemes are required.

B. Literature Survey

Building on the Deep Learning (DL)-based approaches, spectrum sensing techniques have been categorized into five main types: Multilayer Perceptrons (MLPs), Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, combined CNN-LSTM architectures, and Autoencoders (AEs). In the literature there are various methods proposed for detecting the number of primary users. Different sensing techniques vary in terms of time constraints, complex-

COMSNETS'25

2025 17th International Conference on COMmunication Systems & NETworkS (COMSNETS)

Improving VR Performance in WiFi-6 using Deep Reinforcement Learning based Spatial Reuse

Ayush N. Patel, *Student Member, IEEE* and Dhaval K. Patel, *Member, IEEE*
School of Engineering and Applied Science, Ahmedabad University
Ahmedabad, Gujarat, India - 380009
email:{ayush.p3@ahduni.edu.in, dhaval.patel@ahduni.edu.in}

Tworks (COMSNETS) | 979-8-3315-3119-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/COMSNETS63942.2025.10885732

***Abstract*—The growing demand for Virtual Reality (VR) applications requires ultra-low latency and high throughput, posing significant coexistence challenges with traditional WiFi-6 devices. Conventionally used static configurations of Clear Channel Assessment (CCA) threshold and transmit power often lead to hidden terminal issues and inefficient use of available channel resources. This paper proposes a Deep Reinforcement Learning (DRL)-based approach using a Deep Q-Network (DQN) to dynamically control both the CCA threshold and transmit power in real-time, ensuring that VR device meets desired throughput and delay requirements under bursty uplink traffic conditions. Using Network Simulator 3 (ns-3), we simulate a standard-proposed TGax VR gaming scenario involving four Basic Service Sets (BSSs) to evaluate the proposed method. The results demonstrate substantial improvements in VR-specific quality metrics, with our approach achieving a 120%, 100% & 50% enhancement in throughput with data rates 10 Mbps, 30 Mbps & 50 Mbps respectively as compared to conventional fixed CCA sensitivity thresholds and transmit power settings while maintaining the normal devices' experience intact. This study highlights the potential of reinforcement learning techniques in efficiently and fairly managing WiFi channel spatial reuse, addressing the high-performance demands of next-generation wireless applications.**

***Index Terms*—IEEE 802.11ax, Virtual Reality, ns-3, Clear Channel Assessment (CCA) Threshold, Transmit Power, Spatial Reuse, Delay, Throughput, Wi-Fi 6.**

between 25 Mbps to 100 Mbps and user plane latency as low as 4 ms. Wireless Local Area Networks (WLANs) are also evolving to exploit the high capacity of mmW bands through standards like IEEE 802.11ad and IEEE 802.11ay, collectively known as Wireless Gigabit (WiGig) [7].

Unlike traditional applications, XR traffic is highly bursty and characterized by variable frame sizes and inter-frame arrival times due to compression techniques and the complexity of the visual scenes being rendered. The variability in traffic patterns necessitates the development of tailored network protocols and traffic models that can accommodate XR's stringent requirements.

A. Background & Motivation

The motivation for this work stems from the necessity to meet the strict QoS requirements of XR applications given their unique traffic characteristic, which is bursty in nature. In such environments, traditional WLAN protocols may struggle to provide the necessary throughput and latency performance, particularly for applications like VR that require high data rates and low motion-to-photon latency.

The spatial reuse of the channel depends on the medium access control (MAC) protocol used by the AP of a BSS.

Undergraduate Research Programme

Research outcomes from the MICxN Lab

NCC'25

2025 National Conference on Communications (NCC)

Optimizing UAV Deployment for Enhanced Detection Performance in Multi-UAV Cooperative Sensing

Rushi Moliya*, *Student Member, IEEE*, Dhaval K. Patel*, *Senior Member, IEEE*,
Miguel López-Benítez†‡, *Senior Member, IEEE*, Aymen Fakhreddine§, *Member, IEEE*
*School of Engineering and Applied Science, Ahmedabad University, Ahmedabad 380009, India
†Department of Electrical Engineering and Electronics, University of Liverpool, Liverpool L69 3GJ, United Kingdom
‡ARIES Research Centre, Antonio de Nebrija University, 28040 Madrid, Spain
§Institute of Networked and Embedded Systems, University of Klagenfurt, 9020 Klagenfurt, Austria
Emails: {rushi.m, dhaval.patel}@ahduni.edu.in, m.lopez-benitez@liverpool.ac.uk, aymen.fakhreddine@aau.at

/25/$31.00 ©2025 IEEE | DOI: 10.1109/NCC63735.2025.10983909

***Abstract*—In this work, we enhance cooperative sensing capabilities in multi Unmanned Aerial Vehicles (UAVs) systems, with each UAV mounted with a directional antenna to detect multiple targets cooperatively. To improve detection accuracy in dynamic environments with noise fluctuations, we adopt an eigenvalue-based detection technique. We maximize total detection probability by simultaneously optimizing the spatial deployment of UAVs and their antenna orientations. To address the inherent non-convexity of this problem, we propose an iterative Particle Swarm Optimization (PSO)-based approach with a penalty method for its fitness function. The proposed approach effectively navigates complex search spaces, managing spatial and antenna constraints to guide solutions toward global optima. Monte Carlo simulations demonstrate that our PSO-based algorithm outperforms current techniques, achieving superior detection performance and robust sensing capabilities. Notably, the proposed scheme achieves a 7.94% improvement with 2 UAVs and a 7.21% improvement with 3 UAVs over the alternating direction penalty method (ADPM)-based scheme, highlighting its effectiveness even with fewer UAVs deployed for cooperative sensing.**

***Index Terms*—Unmanned Aerial Vehicles, Eigenvalue-Based Detection, Particle Swarm Optimization, Cooperative Sensing.**

In [11], authors introduced resource coordination strategies for reliable detection in multi-UAV networks.

However, energy detection depends on accurate knowledge of noise power. Inaccurate estimation of noise power can result in a high probability of false alarms and a signal-to-noise ratio (SNR) wall, making this method highly sensitive to noise fluctuations [12], [13]. This sensitivity leads to performance degradation in UAV-based applications, where dynamic and unpredictable conditions lead to frequent noise variations, especially in low SNR environments.

To overcome the limitations of energy detection identified in prior work [11], we adopt an eigenvalue-based detection (EBD) technique to enhance sensing performance and remains robust under varying noise conditions. Unlike energy detection, EBD does not rely on prior knowledge of noise variance, making it particularly effective under noise uncertainty—an essential advantage for UAV sensing applications [14], [15]. By jointly optimizing UAV deployment and directional antenna configurations, our method ensures better resource coordination

NCC'25

2025 National Conference on Communications (NCC)

Capacity Analysis of Vehicular Networks in a Multi-User Mixed Traffic Scenario

Preet Patel*, *Student Member, IEEE*, Dhaval K. Patel*, *Senior Member, IEEE*
*School of Engineering and Applied Science, Ahmedabad University, Ahmedabad 380009, India
Email: {preet.p3, dhaval.patel}@ahduni.edu.in

DOI: 10.1109/NCC63735.2025.10983153

***Abstract*—The majority of vehicular network studies rely on the idealized assumption of every vehicle being connected and autonomous. The coexistence of CAVs and Regular Vehicles (RVs) has remained an under-explored area and shown new challenges in understanding the communication capacity required to support efficient vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication. In this paper, we present a capacity analysis of cooperative vehicular networks (VNs) in mixed traffic comprising multiple users with download requests, transmitters, and RVs. We study the impact of RVs on the capacity of vehicular networks (VNs). We also perform Monte Carlo simulation that shows the effectiveness of the proposed analytical framework. The results show that the capacity highly depends on the proportion of vehicles, density, inter-infrastructure distance, and sensing range of transmitters. The results can guide the optimum deployment of infrastructures and set the network configurations and traffic policies to ensure reliable communication.**

***Index Terms*—Communication Capacity, Vehicular Networks, Mixed Traffic, vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I).**

been work done on the capacity analysis of VNs in a scenario that takes RVs into account, they consider a single-user scenario having only a single CAV with a download request, making the scenario overly simplified and unrealistic [8].

Studies have also shown that the inclusion of a dynamic speed-changing model based on the time-varying acceleration or the speed model such as the Gaussian Speed Model has almost no effect on the communication capacity. Increasing the number of hops may improve the capacity marginally which is not significant enough and it increases the communication delay [8]. In this work, we show that the capacity depends on the proportion of vehicles, vehicle density, data rate of V2I communications, inter-infrastructure distance, and the sensing range of the transmitters.

To address these limitations, we perform a more realistic capacity analysis by introducing the RVs into the scenario comprising multiple users. Based on that, we establish a

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Research outcomes from the MICxN Lab

COMSNETS'26

Proactive Self-Aware Multi-Agent Reinforcement Learning for Resource Management in NR-V2X Platooning

Raj Koticha*, Dhaval K. Patel†, Kirtan Kalaria‡, Harsh Patel§
*†‡School of Engineering and Applied Science, Ahmedabad University, Ahmedabad 380009, India
§ Techsture Technologies India Private Limited, Ahmedabad 380054, India
Emails: {raj.k1, dhaval.patel, kirtan.k}@ahduni.edu.in, harsh@techsture.com

***Abstract*—This paper investigates the problem of resource management in NR-V2X (New Radio Vehicle-to-Everything) platooning, addressing the critical trade-off between Age of Information (AoI), Cooperative Awareness Messages (CAMs) demand and finite on-board energy reserve that traditional methods ignore. We propose a novel, Self-sufficient, Energy-Aware, Multi-agent Deep Deterministic Policy Gradient (SEAM-DDPG). By allowing platoons to enter low-power sleep mode strategically to conserve energy for future transmissions. To enhance the robustness of agents under the Centralized Training, Decentralized Execution (CTDE) paradigm, we transform agents from being reactive to proactive by incorporating a one-step forward model and imagination-augmented policy. Our simulation results demonstrate that SEAM-DDPG significantly prolongs the operational life of platoons, while maintaining communication performance. Specifically, SEAM-DDPG (Proactive) achieves a remarkable 94.59% improvement in average energy reserve and a 42.83% increase in operational duration compared to our baselines. This achieves significant 568.39% increase in final**

of Information (AoI), which measures the timeliness of these communications [6].

Although significant research has been done in optimizing AoI and CAM delivery via Deep Reinforcement Learning (DRL) / MARL in vehicular platooning, these overlook the finite onboard energy reserve of the platoons [7]–[10]. The recent works that have attempted to jointly optimize AoI and energy consumption primarily consider single-agent or single-vehicle scenarios where there is no multi-vehicle cooperation required [11], [12]. This introduces a complex, multi-objective optimization problem where the goals of communication performance and energy efficiency in multi-platoon scenario are in direct conflict.

Furthermore, reactive policies that make decisions based solely on current observations can be short-sighted, leading to inefficient energy usage and suboptimal coordination in multi-

COMSNETS'26

Curiosity-Driven Deep-RL for Adaptive Handover Optimization in O-RAN on the NH-47 Corridor

Devashree K. Parikh, Dhaval K. Patel, Kashish D. Shah, Harsh Patel
School of Engineering and Applied Science, Ahmedabad University, Ahmedabad, Gujarat, India - 380009
email: {devashree.p@ahduni.edu.in, dhaval.patel@ahduni.edu.in, kashish.s2@ahduni.edu.in,harsh@techsture.com}

***Abstract*—In recent years, O-RAN has emerged as a major shift in cellular networks, replacing proprietary solutions with open, interoperable, and software-driven designs. It facilitates the integration of AI/ML frameworks to enable adaptive and intelligent network management using Near-RT and Non-RT RIC. One of the critical challenges in such networks is handover management. The traditional methods rely on event-based triggers and limited metrics through hard-coded policies, which often struggle to adapt to dynamic user mobility scenarios. The lack of adaptive policy mechanisms and the dependency on static configurations often hinder seamless mobility management in O-RAN. To overcome these limitations, this paper proposes a Curiosity-Driven Deep Reinforcement Learning framework named C-DDQN for O-RAN-compliant handover control. The agent receives intrinsic rewards based on its exploration by using the Intrinsic Curiosity Module (ICM) integrated with the Double Deep Q-Network (DDQN). The proposed method achieves higher data rates while simultaneously reducing block**

inefficient resource utilization, degraded user experience, and frequent ping-pong effects. Moreover, they typically overlook other important performance metrics such as load balancing, block rate, and handover success rate. In 5G and beyond, the network densification, higher carrier frequencies, and stringent URLLC requirements demand more flexible and intelligent solutions, highlighting the limitations of traditional approaches. Artificial intelligence (AI) and Machine Learning (ML) are now generating a lot of buzz in the field of mobility management because of their capacity to model large and complex environments and adapt dynamically at a manageable computation cost [2]. O-RAN enables the integration of these techniques for effective and adaptive optimization through the RAN Intelligent Controller (RIC).

A. Background and Motivation

Undergraduate Research Programme

Research outcomes from the MICxN Lab

Journals & Magazines > IEEE Access > Volume: 10

PU-DetNet: Deep Unfolding Aided Smart Sensing Framework for Cognitive Radio

Publisher: IEEE Cite This PDF

Brijesh Soni ; Dhaval K. Patel ; Sanket B. Shah ; Miguel López-Benítez ; Siddhartan Govindasamy **All Authors**

1	335
Paper Citation	Full Text Views

Undergraduate Research Programme

Research outcomes from the MICxN Lab

Journals & Magazines > IEEE Transactions on Vehicula... > Early Access

Optimal 3GPP Fairness Parameters in 5G NR Unlicensed (NR-U) and WiFi Coexistence

Publisher: IEEE Cite This PDF

Yashraj Kakkad ; Dhaval K Patel ; Sagar Kavaiya ; Sumei Sun ; Miguel López-Benítez **All Authors**

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Undergraduate Research Programme

Awards





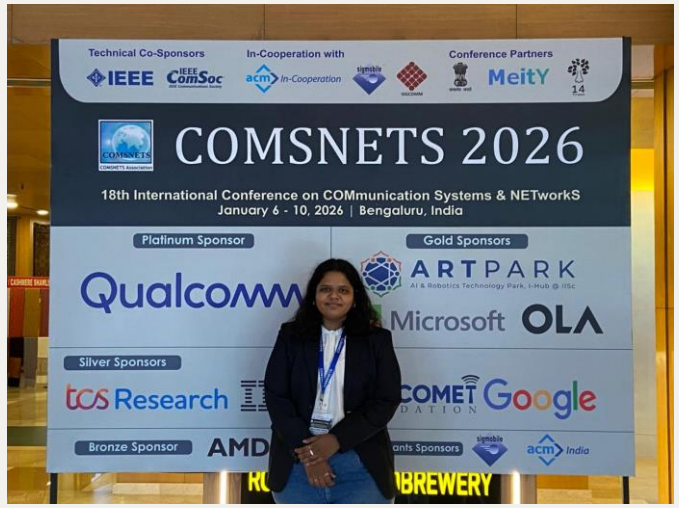




Undergraduate Research Programme

Awards









Ahmedabad University

Undergraduate Research Programme

NCC







MICxN Research Lab – Current Students



Dr. Dhaval Patel
Lab Director

UG Thesis Scholars



Prapti Patel

Advanced Spectrum Sensing



Raj Koticha

O-RAN



Rushi Moliya

6G ISAC + UAV

BTEP Student



Ritu Patel

6G ITS Testbed

MICxN Research Lab – Current Students



Siya Dhaduk

Wi-Fi Sensing



Kaushika Rathod

Wireless Networks



Charmi Goyal

Wireless Networks



Hiya Shah

Wi-Fi 8



Ura Modi

Pinching Antennas



Devashree Parikh

O-RAN



Deep Patel

Pinching Antennas



Shayan Pariyal

Pinching Antennas

Q & A Session: Open Discussions

Thank you!

dhaval.patel@ahduni.edu.in

Thank You!